SSD1306 0.96" I²C OLED Display Module

High-Precision Mechanical Dimensional Guide & Multi-View Engineering Specifications

128 × 64 PIXELS | 4-PIN I²C | 3.3V – 5.0V

1. MODULE OVERVIEW & SPECIFICATIONS

The 0.96-inch OLED module is a high-contrast graphic display based on the Solomon Systech SSD1306 controller. Operating over a 2-wire I²C bus, each of the 8,192 organic LEDs emits its own light, eliminating the need for a backlight and offering wide viewing angles (>160°) with minimal power consumption.

Parameter	Specification
Display Technology	Monochrome OLED (White / Blue / Yellow-Blue)
Active Resolution	128 × 64 pixels (0.96" diagonal)
Active Area (Glass)	21.74 mm × 10.86 mm
Pixel Pitch / Size	0.17 mm × 0.17 mm (Pixel Size: 0.15 × 0.15 mm)
Operating Voltage	3.3V to 5.0V DC (Onboard 662K 3.3V regulator)
Current Draw	~15mA typical (all pixels ON ~25mA, standby <10μA)
Communication Bus	I²C Fast Mode (Up to 400 kHz)
Default I²C Address	`0x3C` (Selectable to `0x3D` via rear resistor)

2. PINOUT & INTERCONNECT TABLE

Pin	Name	Type	Description & Raspberry Pi Pico Connection
1	`GND`	Power	System Ground (Connect to Pico `GND`)
2	`VCC`	Power	3.3V to 5V DC Supply (Connect to `3V3` or `VBUS` 5V)
3	`SCL`	Input	I²C Clock (Connect to Pico `GP1` or `GP5`)
4	`SDA`	I/O	I²C Data (Connect to Pico `GP0` or `GP4`)

3. CRITICAL ENCLOSURE & MOUNTING DIMENSIONS

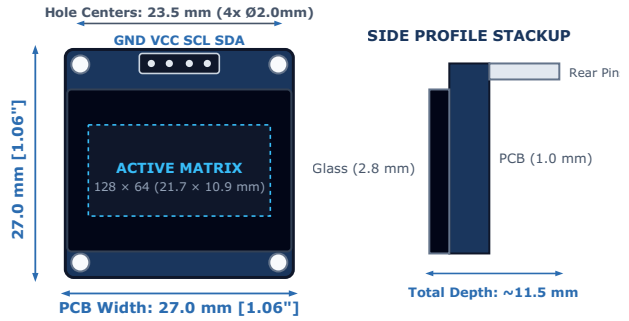


Figure 1: SSD1306 0.96" OLED Mechanical Dimensions, Active Screen Window, and Clearances.

4. ENCLOSURE DESIGN & BEZEL CUTOUT RECOMMENDATIONS

Enclosure Window Sizing: For laser-cut acrylic or 3D-printed bezels, cut a rectangular viewing aperture of **23.0 mm (W) × 12.0 mm (H)**. This fully exposes the active matrix while concealing the outer glass borders. Center the cutout **1.5 mm below** the PCB mounting hole centerline. Use 4× M2 standoffs (minimum 12 mm height if routing straight header DuPont jumpers).

5. CIRCUITPYTHON LIBRARY SETUP

Copy these files to your Pico's `CIRCUITPY/lib/` folder:

- `adafruit_displayio_ssd1306.mpy` (Hardware Driver)
- `adafruit_display_text/` (Folder for Labels & Text)

6. MINIMAL CIRCUITPYTHON CODE EXAMPLE

```
import board, busio, displayio
from i2cdisplaybus import I2CDisplayBus
import adafruit_displayio_ssd1306
from adafruit_display_text import label
import terminalio

displayio.release_displays()
i2c = busio.I2C(board.GP1, board.GP0)
display_bus = I2CDisplayBus(i2c, device_address=0x3C)
display = adafruit_displayio_ssd1306.SSD1306(display_bus, width=128, height=64)

splash = displayio.Group()
text_area = label.Label(terminalio.FONT, text="Hello Maker!\n0.96 OLED", color=0xFFFFFF, x=8, y=16)
splash.append(text_area)
display.root_group = splash
```